AMIRTESH RAGHURAM

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PROFESSIONAL SUMMARY

Computational Biology and Bioinformatics researcher with a Springer Nature publication. Experienced in Computational Drug Discovery, Structure-Based Drug Design (SBDD), Molecular Dynamics simulations, Cheminformatics, and Machine Learning. Proficient in Python, R, Bash, PyTorch, and Next-Generation Sequencing (NGS) analysis. Developer of open-source tools including a PyTorch library with 2,500+ PyPI downloads.

EDUCATION

B.Tech Biotechnology | *Vellore Institute of Technology, Vellore* — 2023 – 2027 | CGPA: 9.01/10

PROFESSIONAL EXPERIENCE

Summer Research Intern | *Strand Life Sciences, Bangalore* — May – July 2025

- Automated Bash pipelines for Next-Generation Sequencing (NGS) data processing, quality control, and clinical genomics variant verification workflows, reducing manual analysis time by ~80%.
- Applied BLAST command-line suite for sequence alignment and homologous region identification across 50+ genomic samples in NGS diagnostic workflows.

PUBLICATIONS

- Raghuram A. et al. "Computational Design and Validation of a Multi-Epitope Subunit Vaccine Against the Neglected Parasite *Blastocystis* sp." *In Silico Pharmacology*, Springer Nature. **Published**, 2026.

RESEARCH PROJECTS

KinaseForge: Generative AI for Kinase Inhibitor Discovery

- Fine-tuned NovoMolGen-157M to generate 2.8M novel SMILES-encoded kinase-like molecules; built XGBoost QSAR models for 25 kinases; deployed searchable multi-target query platform at kinaseforge.vit.ac.in.

Ferroptosis Predictor: Interpretable QSAR Framework

- Classified 1,624 ferroptosis modulators using 2,092 RDKit descriptors + Morgan fingerprints (AUROC 0.9175); SHAP analysis mapped 75 key features to functional group motifs with 86.6% alignment on ChEMBL compounds.

Multi-Target Inhibitors for Triple Negative Breast Cancer (TNBC)

- Screened natural compounds via AutoDock Vina docking, triplicate GROMACS MD simulations, MM/GBSA free energy, and ADMET profiling (SwissADME, pkCSM) for multi-target TNBC lead discovery.

Natural Inhibitors for Glaucoma and Diabetes

- Virtual screening pipelines for Carbonic Anhydrase II (glaucoma) and alpha-amylase (diabetes); validated leads via MD simulations and SwissADME/pkCSM ADMET profiling.

PROJECTS

- **DynaMune** – ENM/NMA protein dynamics platform (ProDy): PRS, domain-hinge and interface stability analysis. GitHub
- **Torchify** – PyTorch utility library; 2,500+ PyPI downloads. GitHub
- **Automated-Virtual-Screening** – Parallel docking pipeline (Vina, Smina, QVina). GitHub
- **Pose-Rescorer** – MM/GBSA rescoring with Rapid Perturbation Sampling; validated on EGFR, HIV-1 PR, BRD4, HSP90. GitHub
- **QUEST** – CLI for ligand strain energies (GFN2-xTB); integrates tautomer enumeration (RDKit), pocket detection (P2Rank), and docking (Vina). GitHub
- **Blood-Brain Barrier Permeability Predictor** – ChemProp v2 MPNN classifier on B3DB (7,807 compounds); MCC-optimized threshold (0.58); metrics >0.9. Live App

TECHNICAL SKILLS

Computational Drug Discovery: Structure-Based Drug Design (SBDD), Molecular Docking (AutoDock Vina, Smina, QVina), GROMACS MD Simulations, MM/GBSA and MM/PBSA Binding Free Energy, ADMET Profiling (SwissADME, pkCSM, ProTox 3.0), In Silico Vaccine Design, ORCA (DFT), xTB and CREST (Conformer/Tautomer Ensembles)

Cheminformatics and AI: RDKit, SMILES, QSAR Modeling, Morgan Fingerprints, Generative Molecular Design, AlphaFold2/3, XGBoost, SHAP Interpretability

Programming and Data Science: Python (NumPy, Pandas, Scikit-learn, BioPython), R, Bash, PyTorch, TensorFlow, Git

Genomics and Next-Generation Sequencing (NGS): RNA-Seq, scRNA-Seq, ChIP-Seq, Variant Calling (GATK), GWAS (PLINK), DNA Methylation, Somatic Mutation Profiling

Structural Bioinformatics: ProDy, Bio3D; PyMOL, UCSF Chimera, Discovery Studio; Protein Modeling (AlphaFold2/3, SWISS-MODEL, I-TASSER)